

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Understanding DATA

```
In [2]: df_cust = pd.read_csv('E:\Work\Forage\Quantium\QVI_purchase_behaviour.csv')
df_cust.head()
```

Out[2]:

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
0	1000	YOUNG SINGLES/COUPLES	Premium
1	1002	YOUNG SINGLES/COUPLES	Mainstream
2	1003	YOUNG FAMILIES	Budget
3	1004	OLDER SINGLES/COUPLES	Mainstream
4	1005	MIDAGE SINGLES/COUPLES	Mainstream

```
In [3]: df_cust.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   LYLTY_CARD_NBR   72637 non-null  int64
1   LIFESTAGE        72637 non-null  object
2   PREMIUM_CUSTOMER 72637 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

Cleaning Data

```
In [4]: df_cust.isna().sum()
```

```
Out[4]: LYLTY_CARD_NBR      0  
        LIFESTAGE          0  
        PREMIUM_CUSTOMER    0  
        dtype: int64
```

```
In [5]: df_cust.duplicated().sum()
```

```
Out[5]: np.int64(0)
```

```
In [6]: df_cust['LIFESTAGE'].value_counts()
```

```
Out[6]: LIFESTAGE  
        RETIREES          14805  
        OLDER SINGLES/COUPLES  14609  
        YOUNG SINGLES/COUPLES  14441  
        OLDER FAMILIES       9780  
        YOUNG FAMILIES       9178  
        MIDAGE SINGLES/COUPLES  7275  
        NEW FAMILIES        2549  
        Name: count, dtype: int64
```

```
In [7]: df_transc =
pd.read_excel('E:\Work\Forage\Quantium\QVI_transaction_data.xlsx')
df_transc.head()
```

Out[7]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PRD
0	43390	1	1000	1	5	Natura Compi SeaSal
1	43599	1	1307	348	66	CCs Na Cheese
2	43605	1	1343	383	61	Smiths Cut Ch Chicke
3	43329	2	2373	974	69	Smiths Thinly S/Crea 175g
4	43330	2	2426	1038	108	Kettle ChpsH Chili 1

```
In [8]: df_transc.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   DATE            264836 non-null int64
1   STORE_NBR       264836 non-null int64
2   LYLTY_CARD_NBR  264836 non-null int64
3   TXN_ID          264836 non-null int64
4   PROD_NBR        264836 non-null int64
5   PROD_NAME       264836 non-null object
6   PROD_QTY        264836 non-null int64
7   TOT_SALES       264836 non-null float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
```

```
In [9]: pd.set_option('display.float_format',lambda x: '%.2f' % x)
```

```
In [10]: df_transc[df_transc.duplicated(keep=False)]
```

Out[10]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	
124843	43374	107	107024	108462	45	
124845	43374	107	107024	108462	45	



```
In [11]: df_transc.drop_duplicates(inplace=True)
```

```
In [12]: df_transc.duplicated().sum()
```

Out[12]: np.int64(0)

RESETTING INDEX

```
In [13]: df_transc.reset_index(drop=True, inplace=True)
```

```
In [14]: df_transc["DATE"].dtype  
df_transc["DATE"].head()
```

Out[14]: 0 43390
1 43599
2 43605
3 43329
4 43330
Name: DATE, dtype: int64

```
In [15]: df_transc['PROD_NAME'].unique()
```

```
Out[15]: array(['Natural Chip          Compny SeaSalt175g',
                'CCs Nacho Cheese      175g',
                'Smiths Crinkle Cut  Chips Chicken 170g',
                'Smiths Chip Thinly  S/Cream&Onion 175g',
                'Kettle Tortilla ChpsHny&Jlpno Chili 150g',
                'Old El Paso Salsa  Dip Tomato Mild 300g',
                'Smiths Crinkle Chips Salt & Vinegar 330g',
                'Grain Waves          Sweet Chilli 210g',
                'Doritos Corn Chip Mexican Jalapeno 150g',
                'Grain Waves Sour    Cream&Chives 210G',
                'Kettle Sensations  Siracha Lime 150g',
                'Twisties Cheese    270g', 'WW Crinkle Cut      Chicken
175g',
                'Thins Chips Light&  Tangy 175g', 'CCs Original 175g',
                'Burger Rings 220g', 'NCC Sour Cream &    Garden Chives 175g',
                'Doritos Corn Chip Southern Chicken 150g',
                'Cheezels Cheese Box 125g', 'Smiths Crinkle      Original
330g',
                'Infzns Crn Crnchers Tangy Gcamole 110g',
                'Kettle Sea Salt    And Vinegar 175g',
                'Smiths Chip Thinly  Cut Original 175g', 'Kettle Original
175g',
                'Red Rock Deli Thai  Chilli&Lime 150g',
                'Pringles Sthrn FriedChicken 134g', 'Pringles Sweet&Spcy BBQ
134g',
                'Red Rock Deli SR    Salsa & Mzzrilla 150g',
                'Thins Chips          Originl salted 175g',
                'Red Rock Deli Sp    Salt & Truffle 150G',
                'Smiths Thinly      Swt Chli&S/Cream175G', 'Kettle Chilli
175g',
                'Doritos Mexicana    170g',
                'Smiths Crinkle Cut  French OnionDip 150g',
                'Natural ChipCo      Hony Soy Chckn175g',
                'Dorito Corn Chp     Supreme 380g', 'Twisties Chicken270g',
                'Smiths Thinly Cut   Roast Chicken 175g',
                'Smiths Crinkle Cut  Tomato Salsa 150g',
                'Kettle Mozzarella  Basil & Pesto 175g',
                'Infuzions Thai SweetChili PotatoMix 110g',
                'Kettle Sensations  Camembert & Fig 150g',
                'Smith Crinkle Cut  Mac N Cheese 150g',
                'Kettle Honey Soy    Chicken 175g',
                'Thins Chips Seasonedchicken 175g',
                'Smiths Crinkle Cut  Salt & Vinegar 170g',
                'Infuzions BBQ Rib   Prawn Crackers 110g',
                'GrnWves Plus Btroot & Chilli Jam 180g',
                'Tyrrells Crisps     Lightly Salted 165g',
                'Kettle Sweet Chilli And Sour Cream 175g',
                'Doritos Salsa      Medium 300g', 'Kettle 135g Swt Pot Sea
Salt',
                'Pringles SourCream  Onion 134g',
                'Doritos Corn Chips  Original 170g',
```

```

'Twisties Cheese      Burger 250g',
'Old El Paso Salsa   Dip Chnky Tom Ht300g',
'Cobs Popd Swt/Chlli &Sr/Cream Chips 110g',
'Woolworths Mild     Salsa 300g',
'Natural Chip Co     Tmato Hrb&Spce 175g',
'Smiths Crinkle Cut  Chips Original 170g',
'Cobs Popd Sea Salt  Chips 110g',
'Smiths Crinkle Cut  Chips Chs&Onion170g',
'French Fries Potato Chips 175g',
'Old El Paso Salsa   Dip Tomato Med 300g',
'Doritos Corn Chips  Cheese Supreme 170g',
'Pringles Original   Crisps 134g',
'RRD Chilli&         Coconut 150g',
'WW Original Corn    Chips 200g',
'Thins Potato Chips  Hot & Spicy 175g',
'Cobs Popd Sour Crm  &Chives Chips 110g',
'Smiths Crnkle Chip  Orgnl Big Bag 380g',
'Doritos Corn Chips  Nacho Cheese 170g',
'Kettle Sensations  BBQ&Maple 150g',
'WW D/Style Chip     Sea Salt 200g',
'Pringles Chicken    Salt Crips 134g',
'WW Original Stacked Chips 160g',
'Smiths Chip Thinly  CutSalt/Vinegr175g', 'Cheezels Cheese
330g',
'Tostitos Lightly    Salted 175g',
'Thins Chips Salt &  Vinegar 175g',
'Smiths Crinkle Cut  Chips Barbecue 170g', 'Cheetos Puffs
165g',
'RRD Sweet Chilli &  Sour Cream 165g',
'WW Crinkle Cut      Original 175g',
'Tostitos Splash Of  Lime 175g', 'Woolworths Medium Salsa
300g',
'Kettle Tortilla ChpsBtroot&Ricotta 150g',
'CCs Tasty Cheese    175g', 'Woolworths Cheese Rings 190g',
'Tostitos Smoked     Chipotle 175g', 'Pringles Barbeque
134g',
'WW Supreme Cheese   Corn Chips 200g',
'Pringles Mystery    Flavour 134g',
'Tyrrells Crisps     Ched & Chives 165g',
'Snbts Whlgrn Crisps Cheddr&Mstrd 90g',
'Cheetos Chs & Bacon Balls 190g', 'Pringles Slt Vingar 134g',
'Infuzions SourCream&Herbs Veg Strws 110g',
'Kettle Tortilla ChpsFeta&Garlic 150g',
'Infuzions Mango     Chutny Papadums 70g',
'RRD Steak &         Chimuchurri 150g',
'RRD Honey Soy       Chicken 165g',
'Sunbites Whlegrn    Crisps Frch/Onin 90g',
'RRD Salt & Vinegar  165g', 'Doritos Cheese Supreme
330g',
'Smiths Crinkle Cut  Snag&Sauce 150g',
'WW Sour Cream &OnionStacked Chips 160g',
'RRD Lime & Pepper    165g',
'Natural ChipCo Sea  Salt & Vinegr 175g',

```

```

'Red Rock Deli Chikn&Garlic Aioli 150g',
'RRD SR Slow Rst      Pork Belly 150g', 'RRD Pc Sea Salt
165g',
'Smith Crinkle Cut    Bolognese 150g', 'Doritos Salsa Mild
300g'],
dtype=object)

```

```

In [16]: # Remove digits from the product names
prod_name = df_transc['PROD_NAME'].str.replace(r'[0-9]+
[gG]', '', regex=True);

# Remove & characters from the product names and replace with a space to
separate flavours
prod_name = prod_name.str.replace(r'&', ' ');
df_transc['PROD_NAME']=prod_name

```

CLEANING CHIPS NAMES

```
In [17]: #correcting chips names
df_transc['PROD_NAME'].unique()
```

```
Out[17]: array(['Natural Chip          Compny SeaSalt', 'CCs Nacho Cheese      ',
                'Smiths Crinkle Cut  Chips Chicken ',
                'Smiths Chip Thinly   S/Cream Onion ',
                'Kettle Tortilla ChpsHny Jlpno Chili ',
                'Old El Paso Salsa   Dip Tomato Mild ',
                'Smiths Crinkle Chips Salt  Vinegar ',
                'Grain Waves          Sweet Chilli ',
                'Doritos Corn Chip Mexican Jalapeno ',
                'Grain Waves Sour     Cream Chives ',
                'Kettle Sensations   Siracha Lime ', 'Twisties Cheese      ',
                'WW Crinkle Cut      Chicken ', 'Thins Chips Light  Tangy ',
                'CCs Original ', 'Burger Rings ',
                'NCC Sour Cream      Garden Chives ',
                'Doritos Corn Chip Southern Chicken ', 'Cheezels Cheese Box ',
                'Smiths Crinkle      Original ',
                'Infzns Crn Crnchers Tangy Gcamole ',
                'Kettle Sea Salt      And Vinegar ',
                'Smiths Chip Thinly  Cut Original ', 'Kettle Original ',
                'Red Rock Deli Thai  Chilli Lime ', 'Pringles Sthrn
FriedChicken ',
                'Pringles Sweet Spcy BBQ ', 'Red Rock Deli SR    Salsa
Mzzrlla ',
                'Thins Chips          Originl salted ',
                'Red Rock Deli Sp    Salt  Truffle ',
                'Smiths Thinly        Swt Chli S/Cream', 'Kettle Chilli ',
                'Doritos Mexicana   ', 'Smiths Crinkle Cut  French OnionDip
                ',
                'Natural ChipCo      Hony Soy Chckn',
                'Dorito Corn Chp    Supreme ', 'Twisties Chicken',
                'Smiths Thinly Cut    Roast Chicken ',
                'Smiths Crinkle Cut    Tomato Salsa ',
                'Kettle Mozzarella    Basil  Pesto ',
                'Infuzions Thai SweetChili PotatoMix ',
                'Kettle Sensations    Camembert  Fig ',
                'Smith Crinkle Cut     Mac N Cheese ',
                'Kettle Honey Soy     Chicken ', 'Thins Chips Seasonedchicken
                ',
                'Smiths Crinkle Cut    Salt  Vinegar ',
                'Infuzions BBQ Rib    Prawn Crackers ',
                'GrnWves Plus Btroot   Chilli Jam ',
                'Tyrrells Crisps      Lightly Salted ',
                'Kettle Sweet Chilli   And Sour Cream ',
                'Doritos Salsa         Medium ', 'Kettle  Swt Pot Sea Salt',
                'Pringles SourCream  Onion ', 'Doritos Corn Chips  Original ',
```



```

'Twisties Cheese      Burger ',
'Old El Paso Salsa   Dip Chnky Tom Ht',
'Cobs Popd Swt/Chlli  Sr/Cream Chips ',
'Woolworths Mild     Salsa ',
'Natural Chip Co     Tmato Hrb Spce ',
'Smiths Crinkle Cut  Chips Original ',
'Cobs Popd Sea Salt  Chips ',
'Smiths Crinkle Cut  Chips Chs Onion',
'French Fries Potato Chips ',
'Old El Paso Salsa   Dip Tomato Med ',
'Doritos Corn Chips  Cheese Supreme ',
'Pringles Original   Crisps ', 'RRD Chilli          Coconut ',
'WW Original Corn    Chips ', 'Thins Potato Chips Hot   Spicy
',
'Cobs Popd Sour Crm   Chives Chips ',
'Smiths Crnkle Chip  Orgnl Big Bag ',
'Doritos Corn Chips  Nacho Cheese ',
'Kettle Sensations   BBQ Maple ', 'WW D/Style Chip      Sea
Salt ',
'Pringles Chicken    Salt Crips ', 'WW Original Stacked Chips
',
'Smiths Chip Thinly  CutSalt/Vinegr', 'Cheezels Cheese ',
'Tostitos Lightly    Salted ', 'Thins Chips Salt   Vinegar ',
'Smiths Crinkle Cut  Chips Barbecue ', 'Cheetos Puffs ',
'RRD Sweet Chilli    Sour Cream ', 'WW Crinkle Cut
Original ',
'Tostitos Splash Of  Lime ', 'Woolworths Medium   Salsa ',
'Kettle Tortilla ChpsBtroot Ricotta ', 'CCs Tasty Cheese   ',
'Woolworths Cheese   Rings ', 'Tostitos Smoked     Chipotle ',
'Pringles Barbeque   ', 'WW Supreme Cheese   Corn Chips ',
'Pringles Mystery     Flavour ',
'Tyrrells Crisps     Ched   Chives ',
'Snbts Whlgrn Crisps Cheddr Mstrd ', 'Cheetos Chs   Bacon
Balls ',
'Pringles Slt Vingar ', 'Infuzions SourCream Herbs Veg Strws
',
'Kettle Tortilla ChpsFeta Garlic ',
'Infuzions Mango     Chutny Papadums ',
'RRD Steak           Chimuchurri ', 'RRD Honey Soy
Chicken ',
'Sunbites Whlegren   Crisps Frch/Onin ', 'RRD Salt   Vinegar
',
'Doritos Cheese       Supreme ', 'Smiths Crinkle Cut  Snag
Sauce ',
'WW Sour Cream  OnionStacked Chips ', 'RRD Lime   Pepper   ',
'Natural ChipCo Sea Salt   Vinegr ',
'Red Rock Deli Chikn Garlic Aioli ',
'RRD SR Slow Rst      Pork Belly ', 'RRD Pc Sea Salt   ',
'Smith Crinkle Cut    Bolognese ', 'Doritos Salsa Mild  '],
dtype=object)

```

```
In [18]: # clearing extra spaces
df_transc['PROD_NAME'] =
df_transc['PROD_NAME'].str.strip().str.replace(r'\s+', ' ', regex=True)
df_transc['PROD_NAME'].unique()
```

```
Out[18]: array(['Natural Chip Compny SeaSalt', 'CCs Nacho Cheese',
                'Smiths Crinkle Cut Chips Chicken',
                'Smiths Chip Thinly S/Cream Onion',
                'Kettle Tortilla ChpsHny Jlpno Chili',
                'Old El Paso Salsa Dip Tomato Mild',
                'Smiths Crinkle Chips Salt Vinegar', 'Grain Waves Sweet
Chilli',
                'Doritos Corn Chip Mexican Jalapeno',
                'Grain Waves Sour Cream Chives', 'Kettle Sensations Siracha
Lime',
                'Twisties Cheese', 'WW Crinkle Cut Chicken',
                'Thins Chips Light Tangy', 'CCs Original', 'Burger Rings',
                'NCC Sour Cream Garden Chives',
                'Doritos Corn Chip Southern Chicken', 'Cheezels Cheese Box',
                'Smiths Crinkle Original', 'Infzns Crn Crnchers Tangy
Gcamole',
                'Kettle Sea Salt And Vinegar', 'Smiths Chip Thinly Cut
Original',
                'Kettle Original', 'Red Rock Deli Thai Chilli Lime',
                'Pringles Sthrn FriedChicken', 'Pringles Sweet Spcy BBQ',
                'Red Rock Deli SR Salsa Mzzrlla', 'Thins Chips Originl salted',
                'Red Rock Deli Sp Salt Truffle', 'Smiths Thinly Swt Chli
S/Cream',
                'Kettle Chilli', 'Doritos Mexicana',
                'Smiths Crinkle Cut French OnionDip',
                'Natural ChipCo Hony Soy Chckn', 'Dorito Corn Chp Supreme',
                'Twisties Chicken', 'Smiths Thinly Cut Roast Chicken',
                'Smiths Crinkle Cut Tomato Salsa', 'Kettle Mozzarella Basil
Pesto',
                'Infuzions Thai SweetChili PotatoMix',
                'Kettle Sensations Camembert Fig',
                'Smith Crinkle Cut Mac N Cheese', 'Kettle Honey Soy Chicken',
                'Thins Chips Seasonedchicken', 'Smiths Crinkle Cut Salt
Vinegar',
                'Infuzions BBQ Rib Prawn Crackers',
                'GrnWves Plus Btroot Chilli Jam', 'Tyrrells Crisps Lightly
Salted',
                'Kettle Sweet Chilli And Sour Cream', 'Doritos Salsa Medium',
                'Kettle Swt Pot Sea Salt', 'Pringles SourCream Onion',
                'Doritos Corn Chips Original', 'Twisties Cheese Burger',
                'Old El Paso Salsa Dip Chnky Tom Ht',
                'Cobs Popd Swt/Chlli Sr/Cream Chips', 'Woolworths Mild Salsa',
                'Natural Chip Co Tmato Hrb Spce',
                'Smiths Crinkle Cut Chips Original', 'Cobs Popd Sea Salt
Chips',
                'Smiths Crinkle Cut Chips Chs Onion', 'French Fries Potato
Chips',
                'Old El Paso Salsa Dip Tomato Med',
```

```

'Doritos Corn Chips Cheese Supreme', 'Pringles Original
Crisps',
'RRD Chilli Coconut', 'WW Original Corn Chips',
'Thins Potato Chips Hot Spicy', 'Cobs Popd Sour Crm Chives
Chips',
'Smiths Crinkle Chip Orgnl Big Bag',
'Doritos Corn Chips Nacho Cheese', 'Kettle Sensations BBQ
Maple',
'WW D/Style Chip Sea Salt', 'Pringles Chicken Salt Crisps',
'WW Original Stacked Chips', 'Smiths Chip Thinly
CutSalt/Vinegr',
'Cheezels Cheese', 'Tostitos Lightly Salted',
'Thins Chips Salt Vinegar', 'Smiths Crinkle Cut Chips
Barbecue',
'Cheetos Puffs', 'RRD Sweet Chilli Sour Cream',
'WW Crinkle Cut Original', 'Tostitos Splash Of Lime',
'Woolworths Medium Salsa', 'Kettle Tortilla ChpsBtroot
Ricotta',
'CCs Tasty Cheese', 'Woolworths Cheese Rings',
'Tostitos Smoked Chipotle', 'Pringles Barbeque',
'WW Supreme Cheese Corn Chips', 'Pringles Mystery Flavour',
'Tyrrells Crisps Ched Chives', 'Snbts Whlgrn Crisps Cheddar
Mstrd',
'Cheetos Chs Bacon Balls', 'Pringles Slt Vingar',
'Infuzions SourCream Herbs Veg Strws',
'Kettle Tortilla ChpsFeta Garlic',
'Infuzions Mango Chutny Papadums', 'RRD Steak Chimuchurri',
'RRD Honey Soy Chicken', 'Sunbites Whlegrn Crisps Frch/Onin',
'RRD Salt Vinegar', 'Doritos Cheese Supreme',
'Smiths Crinkle Cut Snag Sauce',
'WW Sour Cream OnionStacked Chips', 'RRD Lime Pepper',
'Natural ChipCo Sea Salt Vinegr',
'Red Rock Deli Chikn Garlic Aioli', 'RRD SR Slow Rst Pork
Belly',
'RRD Pc Sea Salt', 'Smith Crinkle Cut Bolognese',
'Doritos Salsa Mild'], dtype=object)

```

```
In [19]: # replacing correct names
replacements = {
    'Compny': 'Company',
    'ChipCo': 'Chip Co',
    'SeaSalt': 'Sea Salt',
    'Vinegr': 'Vinegar',
    'Originl': 'Original',
    'Orgnl': 'Original',
    'Slt': 'Salt',
    'Vingar': 'Vinegar',
    'Chimuchurri': 'Chimichurri',
    'Frch/Onin': 'French Onion',
    'Hony': 'Honey',
    'Tmato': 'Tomato',
    'Chckn': 'Chicken',
    'Swt': 'Sweet',
    'Chli': 'Chilli',
    'Btroot': 'Beetroot',
    'Mzzrlla': 'Mozzarella',
    'Gcamole': 'Guacamole',
    'SourCream': 'Sour Cream',
    'Seasonedchicken': 'Seasoned Chicken',
    'FriedChicken': 'Fried Chicken',
    'OnionDip': 'Onion Dip',
    'SweetChili': 'Sweet Chilli',
}

for old_name,new_name in replacements.items():

df_transc['PROD_NAME']=df_transc['PROD_NAME'].str.replace(old_name,new_name
,regex=False)
```

```
In [20]: df_transc['PROD_NAME'] = (
    df_transc['PROD_NAME']
    .str.strip()
    .str.replace(r'\s+', ' ', regex=True)
    .str.title()
)
df_transc['PROD_NAME'].unique()
```

```
Out[20]: array(['Natural Chip Company Sea Salt', 'Ccs Nacho Cheese',
    'Smiths Crinkle Cut Chips Chicken',
    'Smiths Chip Thinly S/Cream Onion',
    'Kettle Tortilla Chpshny Jlpno Chili',
    'Old El Paso Salsa Dip Tomato Mild',
    'Smiths Crinkle Chips Salt Vinegar', 'Grain Waves Sweet
    Chilli',
    'Doritos Corn Chip Mexican Jalapeno',
    'Grain Waves Sour Cream Chives', 'Kettle Sensations Siracha
    Lime',
    'Twisties Cheese', 'Ww Crinkle Cut Chicken',
    'Thins Chips Light Tangy', 'Ccs Original', 'Burger Rings',
    'Ncc Sour Cream Garden Chives',
    'Doritos Corn Chip Southern Chicken', 'Cheezels Cheese Box',
    'Smiths Crinkle Original', 'Infzns Crn Crnchers Tangy
    Guacamole',
    'Kettle Sea Salt And Vinegar', 'Smiths Chip Thinly Cut
    Original',
    'Kettle Original', 'Red Rock Deli Thai Chilli Lime',
    'Pringles Sthrn Fried Chicken', 'Pringles Sweet Spcy Bbq',
    'Red Rock Deli Sr Salsa Mozzarella', 'Thins Chips Original
    Saltd',
    'Red Rock Deli Sp Salt Truffle',
    'Smiths Thinly Sweet Chilli S/Cream', 'Kettle Chilli',
    'Doritos Mexicana', 'Smiths Crinkle Cut French Onion Dip',
    'Natural Chip Co Honey Soy Chicken', 'Dorito Corn Chp
    Supreme',
    'Twisties Chicken', 'Smiths Thinly Cut Roast Chicken',
    'Smiths Crinkle Cut Tomato Salsa', 'Kettle Mozzarella Basil
    Pesto',
    'Infuzions Thai Sweet Chilli Potatomix',
    'Kettle Sensations Camembert Fig',
    'Smith Crinkle Cut Mac N Cheese', 'Kettle Honey Soy Chicken',
    'Thins Chips Seasoned Chicken', 'Smiths Crinkle Cut Salt
    Vinegar',
    'Infuzions Bbq Rib Prawn Crackers',
    'Grnwves Plus Beetroot Chilli Jam',
    'Tyrrells Crisps Lightly Salted',
    'Kettle Sweet Chilli And Sour Cream', 'Doritos Salsa Medium',
    'Kettle Sweet Pot Sea Salt', 'Pringles Sour Cream Onion',
    'Doritos Corn Chips Original', 'Twisties Cheese Burger',
    'Old El Paso Salsa Dip Chnky Tom Ht',
    'Cobs Popd Sweet/Chlli Sr/Cream Chips', 'Woolworths Mild
    Salsa',
    'Natural Chip Co Tomato Hrb Spce',
```

```

'Smiths Crinkle Cut Chips Original', 'Cobs Popd Sea Salt
Chips',
'Smiths Crinkle Cut Chips Chs Onion', 'French Fries Potato
Chips',
'Old El Paso Salsa Dip Tomato Med',
'Doritos Corn Chips Cheese Supreme', 'Pringles Original
Crisps',
'Rrd Chilli Coconut', 'Ww Original Corn Chips',
'Thins Potato Chips Hot Spicy', 'Cobs Popd Sour Crm Chives
Chips',
'Smiths Crnkle Chip Original Big Bag',
'Doritos Corn Chips Nacho Cheese', 'Kettle Sensations Bbq
Maple',
'Ww D/Style Chip Sea Salt', 'Pringles Chicken Salt Crisps',
'Ww Original Stacked Chips', 'Smiths Chip Thinly
Cutsalt/Vinegar',
'Cheezels Cheese', 'Tostitos Lightly Salted',
'Thins Chips Salt Vinegar', 'Smiths Crinkle Cut Chips
Barbecue',
'Cheetos Puffs', 'Rrd Sweet Chilli Sour Cream',
'Ww Crinkle Cut Original', 'Tostitos Splash Of Lime',
'Woolworths Medium Salsa', 'Kettle Tortilla Chpsbeetroot
Ricotta',
'Ccs Tasty Cheese', 'Woolworths Cheese Rings',
'Tostitos Smoked Chipotle', 'Pringles Barbeque',
'Ww Supreme Cheese Corn Chips', 'Pringles Mystery Flavour',
'Tyrrells Crisps Ched Chives', 'Snbts Whlgrn Crisps Cheddar
Mstrd',
'Cheetos Chs Bacon Balls', 'Pringles Salt Vinegar',
'Infuzions Sour Cream Herbs Veg Strws',
'Kettle Tortilla Chpsfeta Garlic',
'Infuzions Mango Chutny Papadums', 'Rrd Steak Chimichurri',
'Rrd Honey Soy Chicken', 'Sunbites Whlegrn Crisps French
Onion',
'Rrd Salt Vinegar', 'Doritos Cheese Supreme',
'Smiths Crinkle Cut Snag Sauce',
'Ww Sour Cream Onionstacked Chips', 'Rrd Lime Pepper',
'Natural Chip Co Sea Salt Vinegar',
'Red Rock Deli Chikn Garlic Aioli', 'Rrd Sr Slow Rst Pork
Belly',
'Rrd Pc Sea Salt', 'Smith Crinkle Cut Bolognese',
'Doritos Salsa Mild'], dtype=object)

```

```
In [21]: conditions = [
    df_transc['PROD_NAME'].str.contains('Salsa', case=False, na=False),
    df_transc['PROD_NAME'].str.contains('Crackers', case=False, na=False),
    df_transc['PROD_NAME'].str.contains('Corn', case=False, na=False),
    df_transc['PROD_NAME'].str.contains('Tortilla', case=False, na=False),
]

choices = ['Salsa', 'Crackers', 'Corn Chips', 'Tortilla Chips']

df_transc['PRODUCT_TYPE'] = np.select(
    conditions,
    choices,
    default='Potato Chips'
)
df_transc
```

Out[21]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
0	43390	1	1000	1	5
1	43599	1	1307	348	66
2	43605	1	1343	383	61
3	43329	2	2373	974	69
4	43330	2	2426	1038	108
...	...	...	...	...	...
264830	43533	272	272319	270088	89
264831	43325	272	272358	270154	74
264832	43410	272	272379	270187	51

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
264833	43461	272	272379	270188	42
264834	43365	272	272380	270189	74

264835 rows × 9 columns

```
In [22]: df_transc['BRAND']=df_transc['PROD_NAME'].str.split().str[0]
```



```
In [23]: df_transc1=pd.read_excel('E:\Work\Forage\Quantium\QVI_transaction_data.xlsx')
df_transc['PACK_SIZE(in grams)']=
df_transc1['PROD_NAME'].str.extract(r'([0-9]+)').astype(float)
df_transc
```

Out[23]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
0	43390	1	1000	1	5
1	43599	1	1307	348	66
2	43605	1	1343	383	61
3	43329	2	2373	974	69
4	43330	2	2426	1038	108
...	...	...	...	...	...
264830	43533	272	272319	270088	89
264831	43325	272	272358	270154	74
264832	43410	272	272379	270187	51
264833	43461	272	272379	270188	42
264834	43365	272	272380	270189	74

264835 rows × 11 columns

```
In [24]: most_bought_chips_brand=
df_transc['BRAND'].value_counts().sort_values(ascending=False).head(5)
most_bought_chips_brand
```

```
Out[24]: BRAND
Kettle      41288
Smiths      28859
Pringles    25102
Doritos     24962
Thins       14075
Name: count, dtype: int64
```

```
In [25]: most_bought_chips_size = df_transc['PACK_SIZE(in
grams)'].value_counts().sort_values(ascending=False)
most_bought_chips_size
```

```
Out[25]: PACK_SIZE(in grams)
175.00      66389
150.00      43131
134.00      25102
110.00      22387
170.00      19983
165.00      15297
300.00      15166
330.00      12540
380.00       6418
270.00       6285
210.00       6272
200.00       4473
135.00       3257
250.00       3169
90.00        3008
190.00       2995
160.00       2970
220.00       1564
70.00        1507
180.00       1468
125.00       1454
Name: count, dtype: int64
```

In [26]: df_transc

Out[26]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
0	43390	1	1000	1	5
1	43599	1	1307	348	66
2	43605	1	1343	383	61
3	43329	2	2373	974	69
4	43330	2	2426	1038	108
...	...	...	...	...	...
264830	43533	272	272319	270088	89
264831	43325	272	272358	270154	74
264832	43410	272	272379	270187	51
264833	43461	272	272379	270188	42
264834	43365	272	272380	270189	74

264835 rows × 11 columns



```
In [27]: # df_transc2=df_transc.copy()
# df_cust1=df_cust.copy()
```

```
In [28]: df = pd.merge(df_cust,df_transc,on='LYLTY_CARD_NBR',how='inner')
df.head(5)
```

Out[28]:

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER	DATE
0	1000	YOUNG SINGLES/COUPLES	Premium	43390
1	1002	YOUNG SINGLES/COUPLES	Mainstream	43359
2	1003	YOUNG FAMILIES	Budget	43531
3	1003	YOUNG FAMILIES	Budget	43532
4	1004	OLDER SINGLES/COUPLES	Mainstream	43406



In []:

1. Which customer segment drives the most sales?

```
In [29]: df.groupby('LIFESTAGE')
         ['TOT_SALES'].agg('sum').sort_values(ascending=False)
```

```
Out[29]: LIFESTAGE
         OLDER SINGLES/COUPLES    402420.75
         RETIREES                366470.90
         OLDER FAMILIES          353767.20
         YOUNG FAMILIES          316160.10
         YOUNG SINGLES/COUPLES    260405.30
         MIDAGE SINGLES/COUPLES    184751.30
         NEW FAMILIES             50433.45
         Name: TOT_SALES, dtype: float64
```

```
In [30]: df.groupby('PREMIUM_CUSTOMER')
         ['TOT_SALES'].agg('sum').sort_values(ascending=False)
```

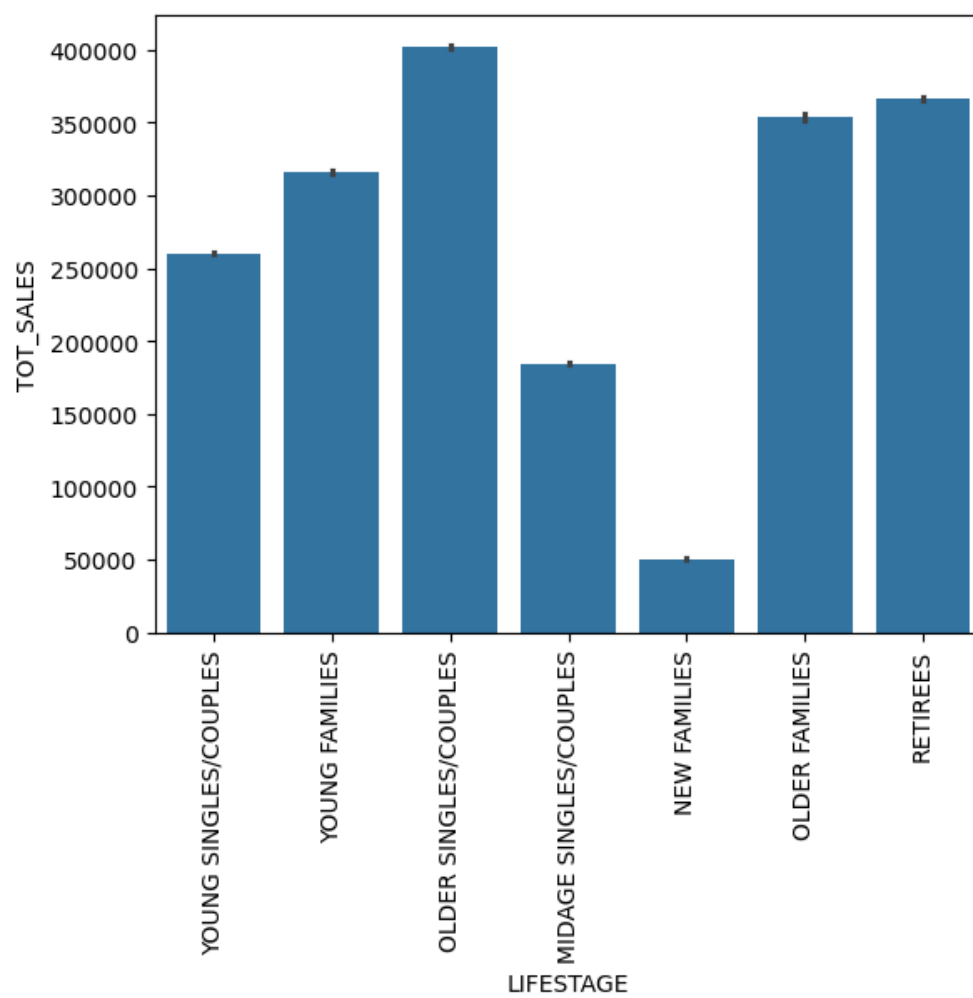
```
Out[30]: PREMIUM_CUSTOMER
         Mainstream    750744.50
         Budget        676211.55
         Premium       507452.95
         Name: TOT_SALES, dtype: float64
```

```
In [31]: df.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'], as_index=False)
['TOT_SALES'].agg('sum').sort_values(by='TOT_SALES', ascending=False).head(10)
```

Out[31]:

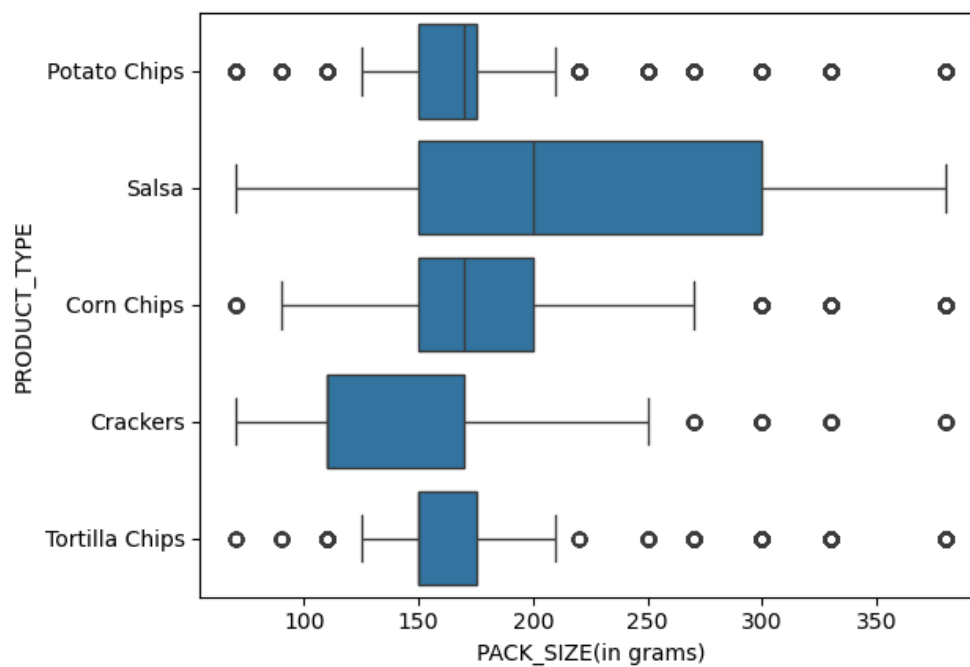
	LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES
6	OLDER FAMILIES	Budget	168363.25
19	YOUNG SINGLES/COUPLES	Mainstream	157621.60
13	RETIREEES	Mainstream	155677.05
15	YOUNG FAMILIES	Budget	139345.85
9	OLDER SINGLES/COUPLES	Budget	136769.80
10	OLDER SINGLES/COUPLES	Mainstream	133393.80
11	OLDER SINGLES/COUPLES	Premium	132257.15
12	RETIREEES	Budget	113147.80
7	OLDER FAMILIES	Mainstream	103445.55
14	RETIREEES	Premium	97646.05

```
In [32]: sns.barplot(data=df,x='LIFESTAGE',y='TOT_SALES',estimator='sum')  
plt.xticks(rotation=90)  
plt.show()
```



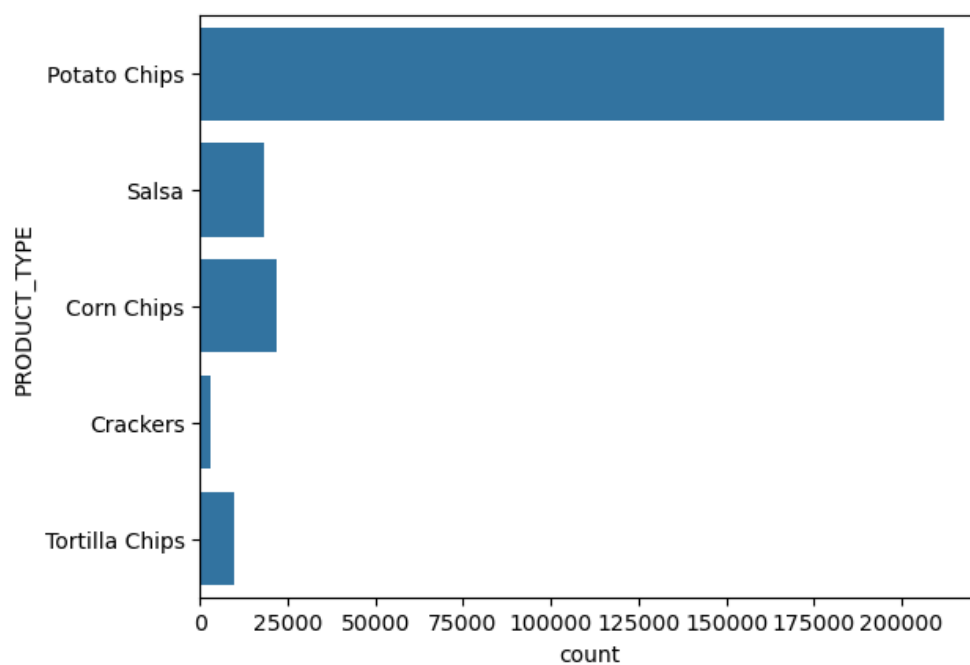
```
In [33]: sns.boxplot(data=df,x='PACK_SIZE(in grams)',y='PRODUCT_TYPE')
```

Out[33]: <Axes: xlabel='PACK_SIZE(in grams)', ylabel='PRODUCT_TYPE'>



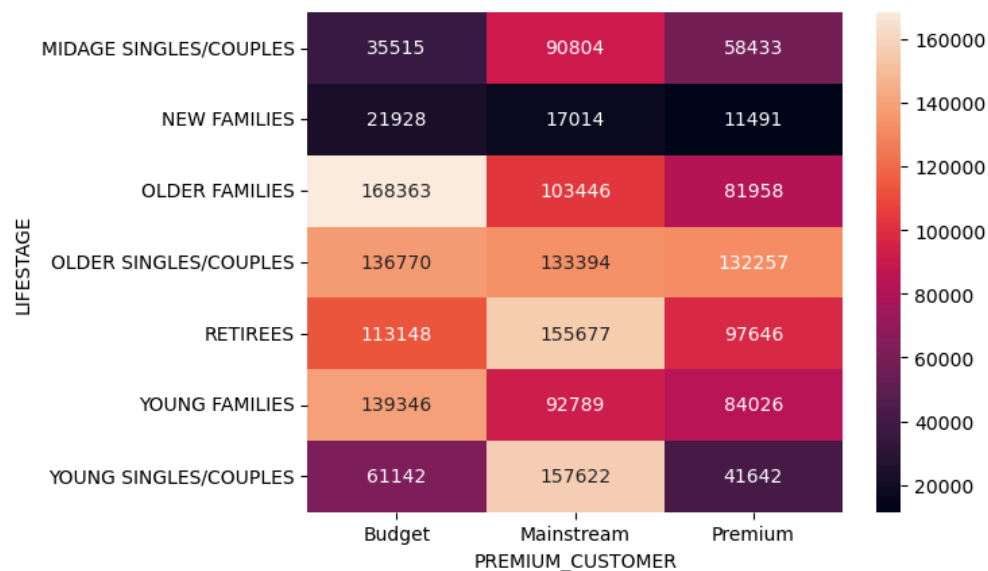
```
In [34]: sns.countplot(  
    data=df,  
    y='PRODUCT_TYPE'  
)
```

Out[34]: <Axes: xlabel='count', ylabel='PRODUCT_TYPE'>



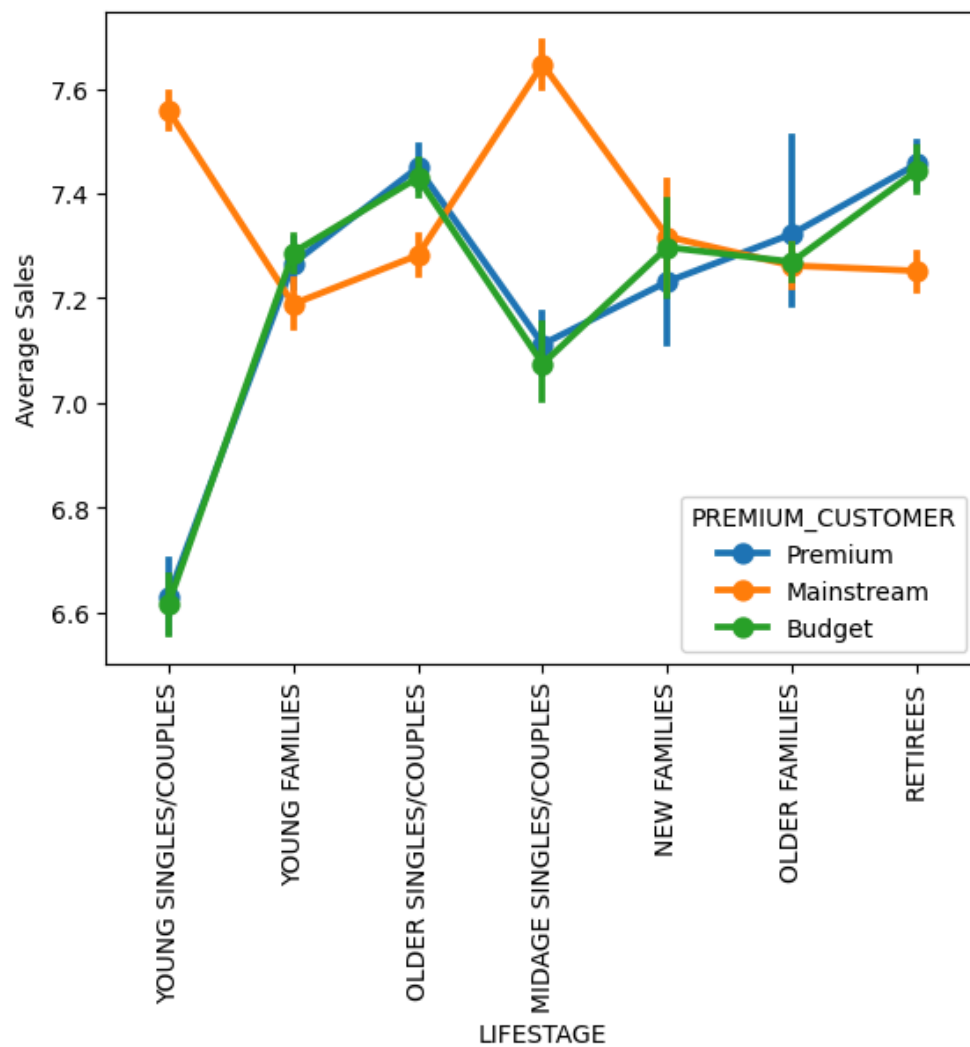

```
In [38]: sales = df.groupby(
            ['LIFESTAGE', 'PREMIUM_CUSTOMER']
        )['TOT_SALES'].sum().unstack()

sns.heatmap(sales, annot=True, fmt='.0f')
plt.show()
```

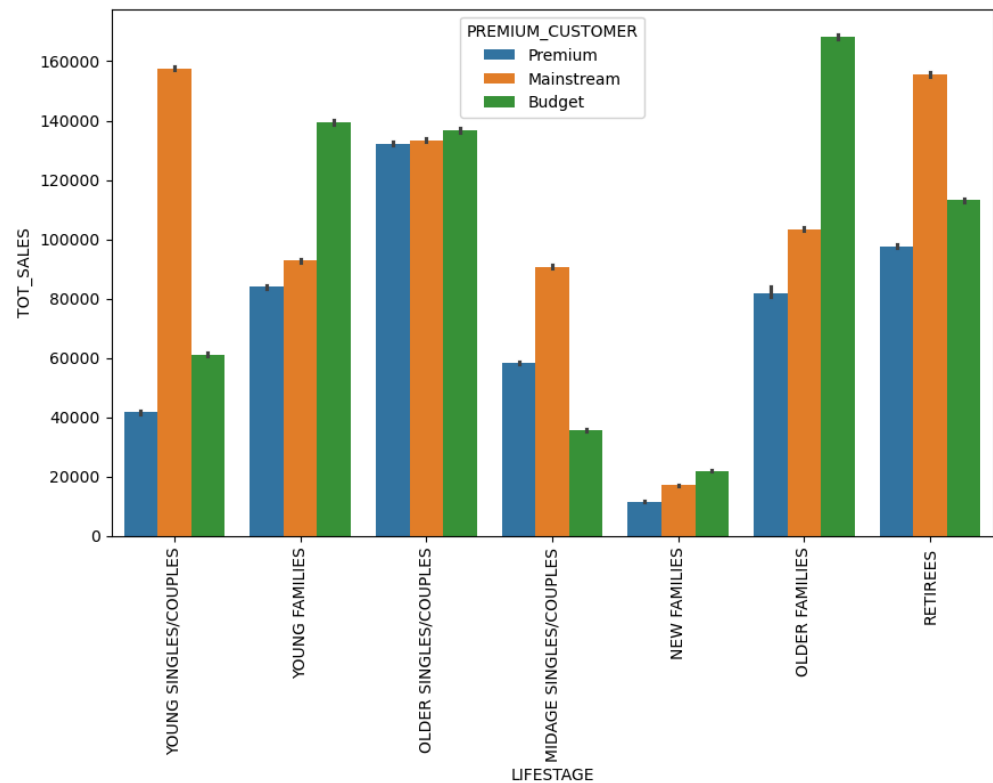


We can see that the sales are higher for the customers who are in the 'Young Singles/Couples' and 'Mainstream', 'Retirees' and 'Mainstream', and for 'Older families' and 'Budget' Premium customers.

```
In [56]: sales_breakdown = df.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'],  
as_index=False)['TOT_SALES'].agg(['sum', 'mean'])  
sales_breakdown  
sns.pointplot(data=df, x='LIFESTAGE', y='TOT_SALES', hue='PREMIUM_CUSTOMER')  
plt.xticks(rotation=90)  
plt.ylabel('Average Sales')  
plt.show()
```



```
In [59]: plt.figure(figsize=(10,6))
sns.barplot(
    data=df,
    x='LIFESTAGE',
    y='TOT_SALES',
    hue='PREMIUM_CUSTOMER',
    estimator=sum
)
plt.xticks(rotation=90)
plt.show()
```



```
In [ ]:
```